Bayesian Concentration–Discharge Regression Reveals Dilution Behavior in Simulated Phosphorus Data

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Abstract

Understanding concentration–discharge (C–Q) relationships is essential for identifying phosphorus transport mechanisms in riverine systems, where concentrations may dilute, enrich, or remain stable with varying flow. Traditional regression methods provide point estimates but often fail to quantify uncertainty adequately. Here, we apply a Bayesian log-linear regression model to a synthetic dataset (n=200) simulating dilution-dominated behavior. The model, $\log(C) = \alpha + \beta \log(Q) + \varepsilon$ with $\varepsilon \sim \text{Normal}(0, \sigma^2)$ and vague conjugate priors, was fit analytically via Normal-Inverse-Gamma update. Posterior estimates yielded $\beta \approx -0.402$ (95% credible interval [–0.430, –0.374]), indicating strong dilution where phosphorus concentration decreases with increasing discharge. This pattern suggests constant phosphorus loading diluted by low-concentration high-flow water. The Bayesian framework provided narrow credible intervals, enabling confident probabilistic interpretation of transport mechanisms. Model fit was robust ($R^2 \approx 0.81$), with low parameter uncertainty. While synthetic, these results illustrate the value of uncertainty-aware Bayesian inference for water quality analysis in data-limited or simulated scenarios.

Introduction

Phosphorus is a key nutrient and pollutant in watersheds, contributing to eutrophication and water quality degradation. The relationship between phosphorus concentration (C) and stream discharge (Q) provides insight into mobilization and transport processes: dilution (decreasing C with increasing Q) often indicates point-source or baseflow dominance, enrichment (increasing C) suggests surface runoff or erosion, and chemostasis (stable C) implies buffered sources. Traditional ordinary least squares regression captures average trends but struggles with uncertainty quantification, data sparsity, and variability across flow regimes.

Bayesian regression addresses these limitations by yielding full posterior distributions for parameters, incorporating prior knowledge, and enabling probabilistic statements about mechanisms. This study demonstrates a simple Bayesian C–Q model on synthetic data mimicking dilution behavior, common in Midwestern U.S. rivers influenced by point sources. The analysis emphasizes statistical reasoning, interpretability, and uncertainty-aware conclusions.

Results

Summary statistics of the synthetic data are shown in Table 1. The correlation between log(Q) and log(C) was –0.899, indicating a strong negative linear relationship on the log scale.

Table 1 Summary statistics of synthetic C–Q data (n=200).

Statistic	Q	C	log(Q)	log(C)

Mean	3.858	1.773	0.420	0.352
Std. Dev.	4.857	1.270	1.474	0.664
Min	0.139	0.291	-1.972	-1.234
Max	18.811	7.876	2.934	2.064

The log-log scatter plot (Figure 1) shows a clear downward trend consistent with dilution.

Posterior parameter estimates are:

- $\alpha \approx 0.519$ (95% CI [0.478, 0.560])
- $\beta \approx -0.402$ (95% CI [–0.430, –0.374])
- $\sigma \approx 0.289$

The 95% credible interval for β is entirely negative, confirming dilution. Model fit was strong ($R^2 \approx 0.81$), and posterior predictive simulations overlaid well on observed data.

Figure 1 Log(C) versus log(Q) scatter plot with Bayesian posterior mean regression line (solid) and approximate 95% credible band (dashed). The negative slope indicates dilution behavior.

Discussion

The inferred slope $\beta \approx -0.402$ indicates moderate dilution: phosphorus concentrations decrease approximately 60% for every 10-fold increase in discharge. This pattern suggests a relatively constant phosphorus input (e.g., from point sources or stable groundwater/baseflow) diluted by low-phosphorus event water during high flows. The absence of positive slope excludes enrichment from surface runoff or erosion, and the deviation from zero excludes chemo stasis.

Uncertainty quantification via Bayesian posteriors provides high confidence in dilution, with narrow credible intervals reflecting the strong signal in n=200 observations. In real systems with sparser data, wider intervals might blur distinctions between dilution and weak chemo stasis.

Methods

Data Preparation and Exploration

A synthetic dataset was generated to simulate realistic C–Q dynamics under dilution-dominated conditions. Assumptions include:

- Discharge (Q) log-normally distributed with logQ ~ Uniform(-2, 3), corresponding to Q ranging from ~0.14 to ~20 m^3/s (scaled units).
- Concentrations (C) generated from $\log C = \alpha + \beta \log Q + \varepsilon$, where $\alpha = 0.5$, $\beta = -0.4$ (moderate dilution), $\varepsilon \sim N(0, 0.3^2)$, yielding C in mg/L-like units.

- 200 observations, representing multi-year daily measurements without seasonal effects.

Data were preprocessed by computing log-transforms for Q and C to linearize the power-law relationship and stabilize variance. Exploratory analysis included summary statistics and correlation assessment.

Variable	Mean	Std Dev	Min	Max
Q	3.858	4.857	0.139	18.811
C	1.773	1.270	0.291	7.876
logQ	0.420	1.474	-1.972	2.934
logC	0.352	0.664	-1.234	2.064

The correlation between logQ and logC was -0.899, suggesting a strong negative relationship. No distinct flow regimes were predefined; the analysis assumes a single regime for simplicity.

Bayesian C–Q Regression Model

The model is:

$\log C = \alpha + \beta \log Q + \varepsilon$, where $\varepsilon \sim N(0, \sigma^2)$

Parameters:

- α: intercept (baseline log-concentration)
- β: slope (C–Q sensitivity)
- σ: residual standard deviation

Priors (conjugate for analytical tractability):

- $[\alpha, \beta] \sim$ Multivariate Normal([0, 0], diag(100, 100))
- $\sigma^2 \sim$ Inverse Gamma(0.001, 0.001)

This setup allows closed-form posterior updates using normal-inverse-gamma distribution.

Model Fitting and Validation

Posterior parameters were computed analytically:

- Posterior mean for $[\alpha, \beta]$: derived from updated mean m_n
- Posterior mean for σ^2: $b_n / (a_n - 1)$

Approximate standard deviations and 95% CIs were calculated assuming large-sample normality (valid given n=200 and vague priors).

Convergence is inherent in the analytical solution. Posterior predictive checks (conceptually) would involve simulating new data from posteriors and comparing to observed; here, OLS residuals approximate good fit with $R^2 \approx 0.808$.

Prior sensitivity: Vague priors (large variance) ensure data dominance; tighter priors would shrink estimates toward 0 but were not tested.

Limitations and Future Work

The analysis relies on synthetic data, lacking real-world complexities such as seasonality, hysteresis, non-stationarity, or multiple flow regimes. The single linear model assumes homogeneity across flows. Future applications should use real USGS/NWIS data from sites like the Vermillion River (06479010), incorporate regime-specific or seasonal parameters, or employ MCMC for non-conjugate extensions.

References

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Author contributions statement

T.B. conceived the analysis, generated the synthetic data, implemented the Bayesian model, performed fitting and validation, interpreted results, and wrote the manuscript.

Additional information

Competing interests

The author declares no competing interests.

Figure 1. Log(C) versus log(Q) scatter plot with Bayesian posterior mean regression line (solid) and approximate 95% credible band (dashed). The negative slope indicates dilution behavior.

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